

C64 Keyboard (Hot Swap) Rev. 1

Bill of Material Rev. 1.0

Pos.	Qty Value	Footprint	Ref.-No.	Comment	Group
1	1 225-2-01-01	2 Layer	_PCB Rev. 1	2 layer, Cu 35 μ , HASL, 385.6mm x 96.4mm, 1.6mm FR4	
2	3 100n	0805	C1, C2, C3	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	SL
3	1 1000u/10V	C30X16L	C4	electrolytic capacitor, axial leads, maximum length 30mm, maximum $\varnothing$ 16mm. E.g. Reichelt AX 105 1000/10	RGB
4	1 22n	0805	C5	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	SL
5	3 100n	0805	C6, C7, C8	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	POR
6	1 LL4148	SOD80C	D1	Diode, SMD	SL
7	2 LL4148	SOD80C	D2	Diode, SMD	SPK
8	1 SN74HC74D	SO-14	IC1	SMD IC, e.G. TI	SL
9	1 74HC123D	SO-16	IC2	SMD IC, e.G. TI	SL
10	1 TL7705ACD	SO8	IC3	SMD IC, e.G. TI	POR
11	1 box pin header, 2x10, 90°	2X10WH	J1	standard 2x10 pin box pin header, e.g. Reichelt.de WSL 20W	
12	1 pin header, 5p/90°, 2.54mm	1X05/90	J2	standard 90° pin header, e.g. Reichelt BKL 10120522	KSW
13	1 pin header, 5p/90°, 2.54mm	1X05/90	J6	DO NOT PLACE	
14	2 KF2510 , 3p/90°, 2.54mm	1X03/90	J3, J8	KF2510 (AliExpress) or Molex KK 2.54 (3p, 90°)	
15	1 pin header, 2p/90°, 2.54mm	1X02/90	J4	standard 90° pin header, e.g. Reichelt BKL 10120522	KSW
16	1 Jumper, 2.54mm		(J4)	standard 2.54mm jumper, only required for short boards	

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17	1	pin header, 4p/90°, 2.54mm	1X04/90	J5	standard 90° pin header, e.g. Reichelt W+P 946-144	RGB
18	1	pin header, 4p/90°, 2.54mm	1X04/90	J7	standard 90° pin header, e.g. Reichelt W+P 946-144	I²C
19	1	PCF8575-MODULE	PCF8575MODU L	M1	Ali Express PCF8575 module. Read module description. Configuration of solder bridges requires	USBKB
20	2	10 way socket for pin header 2.54mm		(M1)	precision sockets, 2.54mm, 1x10, straight. E.g. Reichelt MPE 115-1-010	USBKB
21	1	5 way socket for pin header 2.54mm		(M1)	precision sockets, 2.54mm, 1x5, straight. E.g. Reichelt MPE 115-1-005	USBKB
22	1	5 way pin header, 2.54mm		(M1)	straight, e.g. Reichelt BKL 10120503	USBKB
23	1	PRO_MICRO	PRO_MICRO	M2	Ali Express pro micro module. USB-C connector preferred. Read module description	CTRL
24	2	12 way socket for pin header 2.54mm		(M2)	precision sockets, 2.54mm, 1x12, straight. E.g. Reichelt MPE 115-1-012	CTRL
25	3	BC847B	SOT23	Q1, Q3, Q5		
26	1	100R*	0805	R1	Resistor SMD 0805, 5% or better. Read Module Description	SL
27	1	330R	0805	R11	Resistor SMD 0805, 5% or better	
28	1	470R	0805	R16	Resistor SMD 0805, 5% or better	RGB
29	1	4k7	0805	R18	Resistor SMD 0805, 5% or better	SPK
30	1	33R	0805	R19	Resistor SMD 0805, 5% or better	SPK
31	11	10k	0805	R2, R3, R5, R7	Resistor SMD 0805, 5% or better	SL
32	11	10k	0805	R8, R9, R10, R12, R13, R14, R15	Resistor SMD 0805, 5% or better	
33	1	27k	0805	R20	Resistor SMD 0805, 5% or better	SPK
34	1	27k	0805	R4	Resistor SMD 0805, 5% or better	SL

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35	1 dnp	0805	R6	DO NOT PLACE	
36	1 47k	RN-9	RN1	Resistor network, 8 resistors/9 pins. E.g. USBKB Reichelt SIL 9-8 47K	
37	1 RMP-14P/HT	RMP-14P/HT	SPK1	piezzo buzzer, Ø14mm, pin distance 7.6mm, EKULIT (Reichelt SUMMER EPM 121)	SPK
38	58 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10, SW11, SW12, SW13, SW14, SW15, SW16, SW19, SW20, SW21, SW22, SW23, SW24, SW25, SW26, SW27, SW28, SW29, SW30, SW31, SW34, SW35, SW36, SW37, SW38, SW39, SW40, SW41, SW42, SW43, SW44, SW45, SW46, SW47, SW48, SW50, SW52, SW53, SW54, SW55, SW56, SW57, SW58, SW59, SW60, SW61, SW63, SW64, SW66	Cherry MX compatible switches of your choice. Read the module description.	
39	8 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW17 or SW17A, SW18 or SW18A, SW32 or SW32A, SW33 or SW33A, SW49 or SW49A, SW51 or SW51A, SW62 or SW62A, SW65 or SW65A	SW** = for original keycaps, SW**A = for modern keycaps. Read module description	
40	2 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW67, SW68		KSW and RGB

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41	76	SMD Hot Swap Sockets		SW1-68, SW17A, SW18A, SW32A, SW33A, SW49A, SW51A, SW62A, SW65A	E.g. Kailh CPG151101S11-16 (AliExpress). Read module description	
42	1	LED (high brightness)	LED 3mm	(SW35)	e.G. Kingbright L-7104SRC-J4 (Reichelt LED 3-4500 RT)	SL
43	1	Stabilizer, screw in type 7U		(SW66)	(for 7U space bar or 9U spacebar of C64U/Blingboard64 (CBMStuff.com). Read module description)	
44	1	Stabilizer, screw in type 2U		(SW48)	for RETURN key	
45	58	1U, blank keycaps			for MX compatibel switches, DSA profile. Read module description	
46	8	1.5U, blank keycaps			for MX compatibel switches, DSA profile. Read module description	
47	1	2U, blank keycap		(SW48)	for MX compatibel switches, DSA profile. Read module description	
48	1	7U space bar		(SW66)	for MX compatibel switches, DSA profile. Read module description	
49	25	C 2.2 x 6,5 H			screw, self tapping, DIN 7981	
50	2	C 2.9 x 9,5 H			screw, self tapping, DIN 7981	
51	2	3.2mm washer				
52	1	3D printed keyboard frame			Rev. 1	
53	1	ribbon cable + connector PCB			Project No. 224	
54	1	kernal switcher cable			Drawing 223-3-001-00. Read module description	KSW
55	1	optional WS2812B RGB LED strip			Read module description	RGB
56	1	optional OLED display			Read module description	I²C

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Grp.	SL	Shift Lock Circuit				
	POR	Power on reset				
	RGB	RGB LED strip				
	SPK	Piezo Buzzer				
	KSW	Kernal Switcher				
	USBKB	USB Keyboard				
	I ² C	I ² C (OLED) Display				
	CTRL	Controller/pro micro				